CAPTURE BOARD

The Refinery Is the Reserve - value-capture board

The single best opportunity is not selling intel to a copper operator, it is taking the position ourselves: long Aurubis (DE0006766504) paired against a copper-price or miner proxy to isolate the TC/RC conversion-margin recovery from the metal price. This wins on all three criteria the brief demands. The payer is obvious and reachable because there is no payer at all, just our own book, so there is no procurement cycle, no need for a metals brand-name, and no gatekeeper who can say no. The edge is genuinely differentiated: every sell-side desk is arguing the copper price while we name the conversion node as the binding constraint and tie it to a hard dated signal (2026 benchmark settled at $0/t, spot TC/RC near -$45 to -$60/t). And the first move is cheap: open a small dated, leak-free, scored position with a clear kill-criterion (TC/RC durably back above ~$20/t). It monetizes the exact edge on the exact signal we hold and it extends the scored record that is the only real asset a small independent shop has. The operator retainer to Aurubis and Freeport is the largest nominal prize but the plan's own refutation is right that it is unreachable cold against incumbents like CRU, Wood Mackenzie and Fastmarkets who already sell this read; demote it to a single cheap one-page brief and treat operators and Commerce/CGEP/CSIS as credibility and distribution, not payers.

Domain	Calls	Pursue	Dated
critical minerals midstream conversion chokepoints	6	1	2026-06-19

Do-this-week shortlist

#	CALL	TARGET	FIRST MOVE	EXPECTED VALUE	EFFORT
1	**Take the position ourselves: long Aurubis vs a copper-price/miner proxy to isolate the TC/RC conversion-margin recovery, logged dated and leak-free as a scored call.**	Our own book (instrument: Aurubis DE0006766504, paired short a copper-price/miner proxy)	Open a small paper or real position long Aurubis vs a copper-miner/price proxy, write it up as a dated card with P, interval, resolution date and kill-criterion (TC/RC durably back above ~$20/t), and log it leak-free.	Medium prize, high odds of landing. The trade captures the exact edge with near-100% executability. Upside bounded by self-funded size (low-four to low-five-figure notional for a 53-call shop) but it converts edge to dollars and to record with zero gatekeeper.	LOW
2	**Submit a one-page leak-free brief into the copper-smelting policy conversation CGEP/CSIS**	Columbia Center on Global Energy Policy (CGEP) and CSIS copper-smelting policy	Find the named CGEP and CSIS authors publishing on US/allied copper smelting (unknown right now: identify them first via their recent	Low direct dollars, medium strategic value. Near-zero cost, a live deadline creates urgency,	LOW

#	CALL	TARGET	FIRST MOVE	EXPECTED VALUE	EFFORT
	are already running, ahead of the June 30 Commerce report, reframing the gap as refining capacity not ore.	circle, as the warm path to the Commerce/BIS Section 232 copper team	published notes), then send a one-page brief stating the dated $0/t benchmark and negative spot TC/RC as proof the constraint is conversion capacity, not ore.	and it builds the credibility layer that any later paid work depends on. Treat as marketing, not a sale.	
3	**Offer one paid standalone note to a commodity-focused fund or base-metals trading desk: the dated TC/RC structural-vs-cyclical call framed as long-the-spread, decorrelated from the China-demand price drag, with our own position attached as conviction.**	A commodity hedge fund or base-metals trading desk that buys decorrelated structural reads (Goehring & Rozencwajg-type is the warm relationship, not the invoice)	Send Adam Rozencwajg (G&R, publicly the front-facing PM, reachable via their open commentary/Seeking Alpha) the dated TC/RC call plus the framing that the edge is the spread not the metal, with our own position attached. Pitch a single paid note to a separate base-metals book in parallel.	Low-to-medium prize, low-to-medium odds. A standalone note is reachable cold at a modest fee; the G&R relationship is high-value but unlikely to convert to cash quickly. Bank nothing on it; pursue it as relationship plus optional one-off.	MEDIUM
4	**Pitch Aurubis corporate strategy a paid diagnostic-then-retainer on whether negative TC/RCs and Western conversion scarcity are structural, timed to their end-of-2026 US smelter go/no-go.**	Aurubis AG, Office of the CEO / Head of Corporate Strategy (CEO Toralf Haag); same play applies to Freeport-McMoRan VP Strategy / IR	Identify the named Aurubis Head of Corporate Strategy (unknown right now: finding the name is the first move) and send a one-page brief stating the dated call and asking one sharp question: does the end-of-2026 US smelter case treat the negative TC/RC as cyclical or structural. Treat any reply as credibility, not a sale.	Large nominal prize, low odds cold. Worth a single cheap brief because the upside is real and the cost is one page, but do not build the plan around it. Operators are distribution and validation, not the payer on this timeline.	MEDIUM

Next 7 days

1. Open the position: long Aurubis (DE0006766504) vs a copper-miner or copper-price proxy, sized to conviction (low-four to low-five-figure notional), and log it as a dated card with P, interval, resolution date, and the kill-criterion that TC/RC durably back above ~$20/t refutes it. This is the base-case dollar and needs no one's permission.

2. Write the one-page leak-free brief: the dated call (2026 benchmark $0/t, spot TC/RC -$45 to -$60/t), the reframe that the constraint is refining capacity not ore, and the named exposed parties (Antofagasta, Aurubis, Freeport/ASARCO). This one page feeds every other move.

3. Find the names. Unknown right now and worth resolving first: the Aurubis Head of Corporate Strategy, and the current CGEP and CSIS authors publishing on US/allied copper smelting. Pull them from recent Aurubis governance/press contacts and the two institutions' recent published notes.

4. Send the brief into the CGEP/CSIS policy circle ahead of the June 30 Commerce report, since the deadline is the urgency and this is the warm path to the BIS Section 232 copper team. Pure credibility play, near-zero cost.

5. Email Adam Rozencwajg at G&R (front-facing PM, reachable via their open commentary) the dated TC/RC call with the spread-not-the-metal framing that directly answers their near-term copper bearishness, with our own position attached as conviction. Pitch a single paid standalone note in the same week to one separate base-metals book.

Copper's binding constraint is moving off the ore and onto non-China smelting capacity; zero/negative TC/RCs are the dated tell, and the buyers are the handful of operators and policymakers deciding right now whether to build refineries in the West.

PURSUE

WHY CAPTURABLE

This is capturable now because the constraint is already showing up as a hard, dated price signal (2026 benchmark settled at $0/t, spot TC/RCs near -$45 to -$60/t) while equity and macro coverage still narrate an ore-scarcity supercycle. That gap is exactly our edge: we are calling that the rent migrates to the conversion node, not the mine. Three reachable payer types exist. First, Western smelter operators (Aurubis, Freeport) are making capital decisions on new US/allied refining capacity literally this year, and their go/no-go hinges on whether negative TC/RCs are cyclical or structural, which is the precise question we answer. Second, the US government is on a dated clock: Commerce must report on domestic copper markets by June 30, 2026 to decide a refined-copper phase-in tariff (15% in 2027, 30% in 2028) plus a Defense Production Act domestic-sales requirement, and they need an independent, leak-free read on whether US conversion capacity can actually close the gap. Third, a natural-resource fund can express the smelter-margin-recovery thesis directly. The one caution: G&R turned near-term bearish on copper the metal on Chinese-demand softness, which actually strengthens our differentiated angle, because our call is on the conversion spread and Western capacity scarcity, not the copper price. The cleanest dollar arrives as a paid intel/advisory retainer to operators and policy shops, with a fund position as the optional self-expressed trade.

TARGETS

Aurubis AG Office of the CEO / Head of Corporate Strategy (US smelter decision) - Toralf Haag (CEO)

Why them: Aurubis has publicly said its end-of-2026 decision is between expanding Richmond recycling, a West Coast plant, or a full new US primary smelter, and has already held preliminary talks with the US government on support. Their entire capital case rests on whether negative TC/RCs and Western conversion scarcity are

VALUE MECHANISM

Paid research/advisory retainer is the realest path: a fixed-scope diagnostic brief converting to a quarterly intel retainer for operators (Aurubis, Freeport) and policy shops, priced on the value of de-risking a 9-figure capital decision and a tariff-timing call. Secondary mechanism: a position we take ourselves to express the thesis (long the smelter-margin / TC-RC recovery rather than the copper price).

WHO PAYS

Aurubis and Freeport corporate strategy/IR budgets pay for the diagnostic-then-retainer: a one-month diagnostic in the ~$15-40k range, converting to a quarterly retainer of roughly $10-50k per the high-ticket bespoke-constraint-forecasting band funds and operators already pay. A natural-resource fund (G&R type) pays a smaller research fee or, more likely, the value is in the relationship and a co-expressed trade. Policy shops (CGEP/CSIS) typically pay little but are the credibility on-ramp to the operator and government budgets.

OUR ANGLE

A small independent shop is credible here precisely because we are not talking the copper price like every sell-side desk; we named the conversion node as the binding constraint and tied it to a dated, falsifiable signal. Our edge is leak-free discipline plus a measured dependency-graph read that distinguishes ore scarcity from refinery scarcity, and a scored record proving we calibrate rather than narrate. To an operator deciding a 9-figure build, an independent, falsifiable, structural read is worth more than another bank's price deck.

PROOF TO SHOW

The dated pre-consensus call with its live signal (2026 benchmark $0/t, spot TC/RC -$45 to -$60/t) plus our leak-free Brier-scored record (0.248 to 0.212 calibration) structural. We hand them the dated dependency-graph read and a pre-consensus call that the rent permanently sits on their node, which de-risks a multi-hundred-million-dollar build.

Reach: Aurubis IR and corporate-strategy team are reachable directly; the AGM-2026 governance deck and press contacts name the comms and strategy leads. Warm route: the Columbia CGEP / CSIS copper-smelting policy circle that Aurubis is already engaging on US support. Cold-email angle: lead with our dated $0/t benchmark call and ask whether their build case treats the negative TC/RC as cyclical or structural.

Freeport-McMoRan `VP Strategy / Investor Relations (Miami smelter, El Paso refinery)`

Why them: Freeport is the incumbent US smelter (Miami, AZ) and refiner (El Paso, TX), is scoping capacity expansion, and authored a Commerce filing positioning itself as 'America's Copper Champion'. They are the single biggest beneficiary if Washington decides the constraint is domestic conversion and backs it with tariffs/DPA. They need an independent forecast of where the conversion rent settles to size their expansion and shape their tariff lobbying.

Reach: Freeport IR is directly contactable; their public BIS filing names the smelter-policy team. First move: find the named VP of Strategy behind the 'America's Copper Champion' submission. Cold angle: offer the leak-free scored read that supports their domestic-conversion-scarcity case ahead of the June 30 Commerce report.

US Department of Commerce, Bureau of Industry and Security `Section 232 copper team / Office preparing the June 30 2026 domestic copper markets report`

Why them: Commerce is directed to report on domestic copper markets by June 30, 2026 to inform whether to impose the refined-copper phase-in tariff and DPA domestic-sales requirement. The core unknown they must judge is whether US/allied conversion capacity can close the 57%-import refined-copper gap. Our call reframes the whole question: it is a refinery shortage, not an ore shortage, which directly changes the policy instrument.

Reach: Public-facing via the Section 232 docket and the CSIS/Columbia CGEP policy channel already publishing on protecting US/allied smelting. First move: submit or circulate a one-page leak-free brief into the copper-

and the entity-graph read naming Antofagasta, Aurubis and Freeport/ASARCO as the real exposed parties.

INSTRUMENT

Long Aurubis (DE0006766504) as the cleanest listed pure-play on Western non-China conversion capacity, expressed as a pair long Aurubis versus a copper-miner or copper-price proxy to isolate the TC/RC-margin recovery from the copper price itself. Direction: long the conversion spread / long Aurubis, short the metal proxy. This isolates exactly the rent migration the call predicts.

smelting policy conversation those two institutions are running, which is the warm path to the BIS team.

Goehring & Rozencwajg Associates Portfolio Manager / Head of Research, natural resource fund - Adam Rozencwajg

Why them: They publish deep copper commentary and run the smelter-bottleneck story, but turned near-term bearish on copper the metal on Chinese demand. Our differentiated, dated call is on the conversion spread and Western capacity, not the copper price, which is a cleaner, decorrelated way to be long the same structural thesis without the China-demand drag. They are a natural buyer of a sharp pre-consensus read and a potential counterparty for the trade.

Reach: G&R publish openly and engage with outside research via Seeking Alpha and their commentary; Rozencwajg is publicly the front-facing PM. Cold angle: send the dated TC/RC call plus our framing that the edge is the spread not the metal, which directly addresses their stated near-term bearishness.

THE ASK

To Aurubis and Freeport: "We will give you a standing, leak-free, scored read on whether negative TC/RCs and Western conversion scarcity are structural or cyclical, on the exact horizon your capital decision needs, before the June 30 Commerce report and through the 2027-2028 tariff phase-in. Start with a paid one-month diagnostic brief, then a quarterly retainer tied to the benchmark and tariff calendar." To Commerce/CGEP/CSIS: "Here is a one-page independent brief showing the constraint is refining capacity, not ore, with the dated $0/t benchmark and negative spot TC/RC as the proof. We can supply an ongoing capacity-gap tracker into your June 30 process." To G&R: "Our call is long the conversion spread and Western smelting scarcity, decorrelated from the copper price your near-term bearishness is about; here is the dated thesis and our scored record."

FIRST MOVE (THIS WEEK)

Send Aurubis corporate strategy a one-page brief this week that states the dated call (TC/RC $0/t benchmark, negative spot) and asks one sharp question: does your end-of-2026 US smelter case treat the negative TC/RC as cyclical or structural, because our scored read says structural and we can prove it. Identify and name the Aurubis Head of Corporate Strategy as part of that move.

CHECKPOINTS

By June 30, 2026 the US Commerce domestic-copper-markets report lands; if it explicitly frames the gap as refining/conversion capacity rather than mine supply, our thesis is opening and the policy buyer is live. | By end of 2026 Aurubis announces its US decision (expand Richmond, West Coast plant, or new primary smelter); a primary-smelter or anode-furnace+tankhouse choice confirms operators are pricing structural conversion scarcity. | Through 2026 the spot/benchmark TC/RC stays at or below zero with no rebound; a sustained move back above ~$20/t would signal the constraint is easing and weaken the call.

DISQUALIFIER

Spot and benchmark TC/RCs rebound durably above ~$20/t (the China capacity halt plus demand softness clears the concentrate tightness), or Commerce's June 30 report frames the problem as ore/mine supply and the policy and capital response targets mining rather than refining. Either means the rent is not migrating to the conversion node and we drop it.

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The drafted primary dollar - a paid diagnostic-then-retainer to Aurubis and Freeport - does not arrive. They are multi-billion-dollar producers whose core job is reading TC/RCs, they already pay CRU, Wood Mackenzie and Fastmarkets for the same structural-vs-cyclical smelter-margin read (CRU even launched a weekly TC/RC assessment in 2026), and a 9-figure build is decided by in-house strategy plus tier-1 consultants and incumbent data houses, not by a cold one-pager from an unknown shop with no metals name. Commerce/CGEP/CSIS pay nothing. The retainer is a fantasy on this timeline.

FIX OR KILL

Flip the primary mechanism: stop selling intel to operators and instead TAKE THE POSITION. The plan already specifies it cleanly: long Aurubis (DE0006766504) paired against a copper-price/miner proxy to isolate the TC/RC conversion-margin recovery from the metal price. This needs no buyer to say yes, no procurement cycle, no industry name, and it monetizes the exact edge (the spread, not the metal) on the exact dated signal we hold. Demote the operator retainer to a speculative long-shot, not the base case. Better-fit buyer than any operator: a natural-resource fund that already trades this complex (G&R-type) - but realistically that is a relationship and a co-expressed trade, not a research invoice, so do not bank revenue on it. If we insist on a paid pilot, the only honest first step is a single cheap paid one-off (not a retainer) to a commodity hedge fund or trading desk that buys decorrelated structural reads, priced as a one-time note, with the position on our own book as proof-of-conviction attached. Operators and Commerce are credibility/distribution, not payers - treat them as such.

> HARDENED ASK
>
> Primary: take the position ourselves - long Aurubis vs a copper-price/miner proxy, sized small, to isolate the conversion-margin recovery; log it dated and leak-free as a scored call. Secondary (optional, low-probability): to a commodity-focused fund or trading desk (G&R-type or a base-metals book), offer ONE paid standalone note - the dated TC/RC structural-vs-cyclical call framed as long-the-spread, decorrelated from the China-demand price drag - at a low one-off fee, with our own position attached as conviction. Do NOT lead with an operator retainer.

REALISTIC TICKET

Self-expressed position: a small book position (size to conviction, e.g. low-four to low-five-figure notional given a 53-call shop, not institutional size). Optional fund note: a one-time research fee in the low single-digit thousands, not the $10-50k/quarter operator-retainer band, which is not reachable cold.

The Western antimony chokepoint is permitted primary trioxide roasting, not ore; and the durable demand is datacenter/grid flame-retardant PVC, not ammunition. Two names own the squeeze (UAMY holds the only permitted OECD roasters; Perpetua holds ore with no roasting home), and the consensus is mispricing why.

PASS

WHY CAPTURABLE

There is a real, reachable capture path because the consensus framing is wrong in a tradeable way. The market and the press tie antimony to the defense story (UAMY's $245M DLA contract, Perpetua's Pentagon/munitions backing), but our thesis isolates two non-obvious truths: (1) the binding constraint is permitted, operating primary Sb_2O_3 volatilization-roasting plus the arsenic-offgas abatement that makes a line permittable, not ore reserves; (2) the demand that genuinely cannot be sourced outside China is flame-retardant Sb_2O_3 in datacenter and grid-electrification plastics, a 6.8% CAGR electronics pull that is structural and price-inelastic, versus lumpy government munitions demand. Our data layer is literally blind on roasting capacity (it indexes ore claims like Antimony Flat and Contango Ore, and the dependency/market walks returned blank), which is the proof the chokepoint is unpriced and unmodeled by the obvious players. That gap is exactly what a sharp forecasting shop sells: a differentiated demand-attribution and permittability read to a fund that trades UAMY/PPTA or to an operator sizing its next roasting line. It is capturable now because UAMY is mid-capex (6x expansion finishing end-2025) and Perpetua is choosing a processing partner (Sunshine Silver MOU, INL pilot), so the decisions our read informs are live this quarter.

TARGETS

United States Antimony Corporation (NYSE American: UAMY) CEO / Head of Investor Relations (find named CEO; Gary Evans associated as Executive Chairman as of 2025)

Why them: They own the only permitted operating primary Sb_2O_3 roasters in North America (Thompson Falls MT + Madero Mexico, expanding to 500 t/mo). Their

VALUE MECHANISM

Two real dollar paths, run in parallel. (1) Paid research/advisory: a fixed-fee constraint-and-demand brief to UAMY/Perpetua/Campine (corporate strategy budget), then a retainer for quarterly updates as the capacity and price calls resolve. This is the non-dilutive, fast path that matches our memo'd high-ticket model. (2) A position we take ourselves: long UAMY (permitted operating roaster, the scarcest asset in the chain) as the cleanest single-name expression, optionally paired against an ore-only/no-roasting-home name to isolate the roasting-capacity premium our thesis identifies. The advisory sale is the primary, lower-variance capture; the position is the conviction expression of the same edge.

WHO PAYS

UAMY / Perpetua / Campine corporate strategy or IR pays $15k-$50k for the scoped constraint-and-demand brief plus $5k-$15k/quarter retainer; a resource hedge fund pays $10k-$30k for the dated pair-trade read or carries it as paid intel. If expressed as our own position: long UAMY equity, conviction-sized, defined kill on the price gap.

OUR ANGLE

We are credible here precisely because we separate the chokepoint from the headline. Anyone can say 'antimony is critical and China cut exports.' Our edge is the dependency-graph discipline that isolates the binding layer (permitted, operating primary roasting plus arsenic-offgas abatement, not ore, not recycling, not wet chemistry) and the demand-attribution that says the durable pull is datacenter/grid flame-retardant plastics, not the lumpy defense story everyone is pricing. We prove it the honest way: a dated, sigma-flagged anomaly and a falsifiable forward call with a kill-criterion, scored at resolution. A small independent shop with a leak-free narrative is defense-led ($245M DLA, $106.7M fabric deal). Our read says the larger, more durable, higher-margin pull is datacenter/grid flame-retardant PVC, which reframes their capex priority and their equity story toward a faster-growing, less lumpy demand base. They also need to defend the permittability moat (arsenic offgas abatement) that keeps new entrants out.

Reach: Public IR contact on UAMY investor page; cold email to IR with a one-page demand-attribution read; or via critical-minerals conferences (Fastmarkets Antimony, Benchmark Source events). First move: find the named CEO/IR director on the latest 8-K and investor deck.

Perpetua Resources (NASDAQ: PPTA) `VP Strategy / Head of Antimony Commercialization (find named exec leading the Sunshine Silver MOU and INL pilot)`

Why them: They hold the largest US antimony ore body (Stibnite, ~150M lb) but have NO roasting line, only MOUs. Our thesis says ore without a permitted roasting home is not supply security, so their entire value-realization hinges on solving the exact chokepoint we have mapped. A demand read that proves the commercial (datacenter/grid) market dwarfs the 10% military allocation strengthens the case for a merchant roasting line, not just a defense feedstock deal.

Reach: PPTA IR; or via Idaho National Laboratory / Sunshine Silver partnership contacts. First move: identify the exec named on the Dec 2025 INL pilot announcement and the Sunshine Silver MOU.

A critical-minerals / resources hedge fund (e.g. a fund with public antimony exposure such as those holding UAMY/PPTA, or a generalist commodities desk) `PM / Analyst covering critical minerals and miners`

Why them: They trade UAMY and PPTA on the defense narrative and the China export-control headline. Our edge is the second-derivative: a leak-free, dated demand-attribution model showing the datacenter/grid pull is the durable driver and that permitted roasting (not ore) is the binding constraint, which separates the names that re-rate (permitted roaster) from those that stall (ore-only, no roasting home). That is a clean long/short or pair-trade thesis.

Reach: Find via 13F holders of UAMY/PPTA, or specialist resource funds. Cold outreach with a dated pre-consensus one-pager. First move: pull recent 13F/13D filers on UAMY and PPTA to name 1-2 funds with real positions.

record is exactly who a fund or an operator trusts on a contrarian second-derivative call.

PROOF TO SHOW

Lead with the dated, leak-free pre-consensus call itself: China antimony export value $118M to $72M (2012-2024) flagged at 21.8-sigma by our detector, plus the explicit, falsifiable headline (by 2034 the binding constraint is permitted primary Sb_2O_3 roasting capacity, demand led by datacenter/grid plastics not ammunition) with a kill-criterion. Pair it with the dependency-graph read showing our own data layer is blind on roasting capacity while it indexes ore claims, which is the demonstration that the obvious players are modeling ore, not the chokepoint. Reinforce with our broader leak-free scored record (calibration 0.248 to 0.212) as the credibility that we score our calls forward, not retrofit them.

INSTRUMENT

Long UAMY (NYSE American: UAMY) as the single-name expression of permitted operating Western primary roasting capacity; optionally paired against an ore-only / no-roasting-home name to isolate the roasting-capacity premium. Direction: long the permitted roaster, with a kill if the Western Sb_2O_3 price gap to China collapses or China durably resumes oxide exports.

Campine NV (Euronext Brussels: CAMB) CEO / Head of Antimony Business Unit

Why them: World-leading Western ATO producer (~18,000 t/yr) but recycling/secondary, not primary roasting, and just approved 3rd-gen recycling capex (Dec 2025). Our read tells them whether the marginal Western tonne the market will pay up for is primary roast (which they do not do) versus recycled, and quantifies the datacenter/grid demand that justifies or caps their expansion. Useful as an advisory client and as a short-side reference if the primary-vs-secondary distinction is what the market is missing.

Reach: Campine IR (Euronext-listed, public investor relations). First move: confirm current CEO from the Dec 2025 Euronext announcement and send the demand/permittability read.

THE ASK

To UAMY and Perpetua: a paid 2-to-4 week constraint-and-demand brief that (1) quantifies the datacenter/grid-electrification Sb_2O_3 flame-retardant pull versus the defense pull with dated demand series, (2) maps the permittability chokepoint (which OECD jurisdictions will permit a primary volatilization roaster given arsenic-offgas abatement rules, and what that does to entrant timelines), and (3) gives a falsifiable, Brier-scored 24-month call on Western primary roasting capacity and the Sb_2O_3 price gap. To the fund: the same read packaged as a long-permitted-roaster / cautious-on-ore-only pair thesis with dated checkpoints. Not a generic partnership; a specific scored deliverable they can act on this quarter.

FIRST MOVE (THIS WEEK)

Pull the most recent UAMY and PPTA SEC filings (latest 8-Ks and the Dec 2025 announcements) to name the exact CEO/IR/commercialization decision-maker at each, then send UAMY IR a one-page dated read titled roughly 'The constraint is permitted roasting, and the demand is datacenters not munitions' that leads with the unpriced-chokepoint finding and offers the scoped brief.

CHECKPOINTS

By end-Q1 2026: UAMY confirms whether the Thompson Falls 6x expansion (500 t/mo) actually came online on schedule (end-2025 target). On-time completion validates the permitted-roaster scarcity thesis; slippage strengthens the chokepoint/permittability angle further. | By mid-2026: Perpetua converts (or fails to convert) the Sunshine Silver MOU and INL pilot into a committed primary roasting/refining line. A committed merchant line that targets commercial (not just defense) Sb_2O_3 confirms our demand-attribution read; a defense-only or stalled path confirms ore-without-a-roasting-home risk. | By end-2026: published Sb_2O_3 price series shows whether the Western (North America ~$56-59k/MT) vs China (~$32k/MT) gap persists or compresses. A persistent gap confirms the binding-constraint premium; durable Chinese oxide-export resumption is the disqualifier.

DISQUALIFIER

China durably resumes antimony-oxide exports at scale and the Western/China Sb_2O_3 price gap collapses, OR enough permitted OECD primary roasting capacity comes online (UAMY 500 t/mo plus Perpetua plus others) that roasting stops being the binding constraint. Either kills both the advisory pitch and the long position. Also disqualifying for the demand half: if independent data shows defense, not datacenter/grid plastics, is and remains the dominant non-substitutable pull.

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The primary named payers are not budgeted buyers of this product, and the capture window is closing in real time. UAMY is a sub-$2B small-cap miner that just posted Q1 2026 revenue of $6.8M (a 66% miss) with negative ~$15M EBITDA, burning cash mid-capex; a money-losing junior has no corporate-strategy line for a $15-50k outside demand-attribution brief, and is the least likely org to pay an unknown shop to be told its own defense-led equity story is mispriced. IR at miners produces research, it does not buy it. Perpetua is pre-revenue and conserving cash for a build; same problem, worse. So the plan's own "primary, lower-variance" path (advisory to UAMY/Perpetua/Campine) targets payers who structurally do not buy this. Worse, the plan's disqualifier is already partly firing: as of Q1 2026 the Western/China price gap is narrowing (off ~36% from the July 2025 $59,750 peak, settling $35-45k), Fastmarkets frames 2026 as "ample supply" led by Southeast Asia/Tajikistan/Turkey/Mexico, and flame-retardant demand reads as stable rather than the structural datacenter pull the thesis needs. The "capturable this quarter" claim is backwards: consensus is moving toward the disqualifier, not toward the squeeze.

FIX OR KILL

Drop the operator-advisory path entirely. UAMY and Perpetua are not budgeted intel buyers; that is unfixable at this ticket. The only real payer is a commodities/critical-minerals fund or desk that already trades UAMY/PPTA on the defense headline. Fix the ask: instead of a $15-50k retainer to a miner, sell a one-time dated pair-trade read (long permitted-roaster / cautious ore-only) to a fund as a cheap paid pilot ($3-5k for one scored 24-month call with checkpoints), priced to prove the edge before any retainer. First move is concrete: pull recent 13F/13D filers on UAMY and PPTA to name 1-2 funds with real positions, then cold-pitch the dated, sigma-flagged one-pager. The differentiation (permitted roasting is the binding layer, not ore; demand attribution datacenter vs munitions) must be tested against what a specialist PM already models, so the pilot is also the test of whether the edge is genuinely non-obvious. Run the self-position as the conviction expression but size it small with a hard kill on the price gap, because the current tape (narrowing gap, ample supply) is going against the squeeze and the entry timing is poor right now.

HARDENED ASK

To a named fund holding UAMY or PPTA (identified from 13F/13D filings): a $3-5k one-time, dated, leak-free pair-trade read (long permitted-operating-roaster / cautious ore-only-no-roasting-home) with a falsifiable 24-month call, explicit checkpoints, and a hard kill on Western/China Sb_2O_3 gap collapse. If it lands and a

> checkpoint resolves our way, then propose a quarterly retainer. No operator-advisory pitch to UAMY/Perpetua.

REALISTIC TICKET

$3-5k paid pilot from one resource fund (not $15-50k from a miner). Retainer only after a resolved checkpoint. Self-position: small, conviction-sized long UAMY with a defined kill on the price gap, entered cautiously given the current narrowing-gap tape.

The binding gate on Western heavy-rare-earth is a licensed radioactive-tails sink, not a separation train. Two private operators (EnergySolutions Clive, WCS Andrews) own that sink, and almost no buyer has priced it.

PASS

WHY CAPTURABLE

The constraint is real, already biting, and under-mapped. NRC/IAEA now treat any feed over 0.05% Th/U as nuclear source material, which drags every Western Dy/Tb plant into 10 CFR 40 / source-material licensing and creates a life-of-plant need for a licensed radioactive-residue sink. Only two US commercial sinks can take 11e(2)/source-material tails (EnergySolutions Clive UT, WCS Andrews TX), and WCS's byproduct license is currently restricted to Fernald, Ohio waste, so there is effectively no clean commercial path for REE source-material tails today. Lynas Seadrift TX is stalled and likely dead specifically over residue/wastewater permitting (Dow withdrew the permit support Feb 2025); MP Materials commissions HREE mid-2026 but owns its own lined zero-discharge sink, which is exactly the differentiator the market is not pricing. Our own dependency graph returned no concept match on this needle, which is the tell: this chokepoint is genuinely off-consensus. Capturable now by (a) selling the dependency-graph read as paid intel to a fund with long/short REE exposure, and (b) an advisory retainer to a processor or a sink operator who needs to know whether the residue gate kills or makes their economics. Not capturable as a clean public equity short on the sink itself because both sink operators are private.

TARGETS

EnergySolutions (ECP / TriArtisan owned, Salt Lake City) President & CEO (and head of commercial/business development under him) - Ken Robuck

Why them: They own License UT 2300478 to take uranium/thorium 11e(2) byproduct at Clive, the single clearest commercial sink for REE source-material tails. If the thesis is right, every Western Dy/Tb plant becomes a multi-decade life-of-plant customer for them, and the license is the rent-bearing asset. They likely have not sized the REE-tails demand wave or positioned Clive's license

VALUE MECHANISM

Two real dollar paths. (1) Paid intel: a fixed-fee constraint memo + briefing call to a resources fund desk (the dependency-graph read packaged as a dated pre-consensus call), repeatable as a quarterly intel feed. (2) Advisory retainer: a monthly retainer from EnergySolutions (or a processor) to map and time the tails-sink demand wave and license positioning. The cleaner public instrument is not the sink (both operators are private) but the intra-sector equity pair on the processor side, which we monetize as the intel product rather than taking the position ourselves.

WHO PAYS

A critical-minerals / resources hedge-fund desk pays roughly 15k to 40k for a one-off dated constraint memo + call, scaling to a 5k-15k/month intel feed if it keeps moving their book (DRW-type firms pay for bespoke constraint-forecasting per our GTM memory). EnergySolutions, if engaged, pays an advisory retainer in the 10k-25k/month range to size and time the REE-tails demand against Clive's license. OSC does not pay cash for a briefing but the relationship is the asset (positions us as the independent constraint shop for the next REE loan).

OUR ANGLE

A small independent forecasting shop is credible here precisely because this is a cross-domain seam (nuclear-waste licensing x rare-earth metallurgy x defense procurement) that no single incumbent owns. The processors think in separation chemistry, the waste firms think in nuclear licensing, and the funds think in price floors; nobody is reading the dependency edge between them. Our edge is the leak-free, dated, gated method that isolates the binding constraint before it is priced, plus a data layer that literally flagged its own blindness on this node.

PROOF TO SHOW

conditions and capacity to capture it. We can hand them the demand map and the timing.

Reach: Robuck is named and reachable via EnergySolutions corporate / LinkedIn; warm path through ECP (Energy Capital Partners) infra team who own the company and think in cash-flow terms. Cold-email angle: 'who is your single biggest source-material-tails customer in 2031, and have you priced the REE wave into Clive's license envelope.'

MP Materials (NYSE: MP) `VP/Head of Operations or Strategy at Mountain Pass (find the named owner of the HREE/Dy-Tb circuit and residue management)`

Why them: They commission the only US HREE (Dy/Tb) circuit mid-2026 and already run Mountain Pass as zero-discharge with lined on-site disposal, meaning they may have effectively internalized the very sink that kills competitors. That is a moat the equity story does not yet name. They care because (a) confirming the residue gate is a competitor-killer strengthens their narrative and DoD case, and (b) any change to source-material license scope is existential to their 200 MT/yr Dy/Tb plan.

Reach: Public company, IR reachable directly; the strategy/government-affairs team is engaged with DoD OSC (they took the $150M OSC loan + $400M equity). Lead with the dependency read as a credibility artifact, ask for the residue-circuit owner.

Office of Strategic Capital (OSC), US Dept of War / DoD `Director, Office of Strategic Capital (and the OSC critical-minerals loan officer running the MP and successor REE deals)`

Why them: OSC just deployed $150M to MP for HREE separation and has $100B (OBBBA) loan authority for critical-minerals capacity. They are funding separation while the actual binding constraint (a licensed residue sink) is unfunded and unmapped. If they finance a Dy/Tb plant with no licensed tails path, the asset strands. They need an independent constraint map before the next loan, exactly the kind of pre-consensus structural read we produce.

Reach: OSC publishes RFIs / announcements via businessdefense.gov (MCEIP DPA Title III) and runs public industry days; respond to an OSC RFI or request a briefing. Find the named OSC director and the REE loan officer as a first move.

Lead with the dependency-graph read that returned NO concept match on this needle (our own data layer is blind here, which is the pre-consensus tell), paired with the dated, leak-free scored record of structural calls. Show the timestamped causal chain: NRC/IAEA 0.05% source-material reclassification -> 10 CFR 40 licensing -> life-of-plant residue sink -> only two constrained US sinks -> Lynas Seadrift failure (Feb 2025) as the already-realized confirmation. The dated 'we flagged the sink, not separation, before the loan officers did' framing is the artifact.

INSTRUMENT

No clean public instrument on the sink itself (EnergySolutions and WCS are both private). The expressible trade is an intra-sector equity pair on the processor side: relatively long the processor that owns/has secured a licensed residue sink (MP Materials, NYSE: MP, zero-discharge lined disposal) versus relatively short or underweight Western HREE processors with no secured 11e(2) path (the Lynas Seadrift / no-sink cohort). Value is largest in the intel/advisory relationship, not in us taking the position.

A fund with directional REE exposure (e.g. a critical-minerals / resources hedge fund or the resources desk at a multi-strat such as a DRW-type firm)

Portfolio Manager / Analyst, materials or critical-minerals book

Why them: They are long/short MP, USAR, Lynas and peers on separation-capacity and price-floor narratives, and almost none of them have the tails-sink constraint in their model. The differentiated, tradable read is: processors that own or have secured a licensed residue sink (MP) versus those that do not (Lynas Seadrift, plants relying on third-party 11e(2) capacity). That is an intra-sector pair, not a sector call, which is exactly what a desk pays for.

Reach: Warm intro via the fundraising/quant network already mapped (DRW-type firms pay for bespoke constraint-forecasting per memory); cold path is a one-page dated call with the pair framing sent to a named materials PM.

THE ASK

To the fund desk: 'Buy a dated, leak-free constraint memo plus a 30-minute call. Thesis: the binding gate on Western HREE is a licensed radioactive-tails sink, not separation. We name the only two US sinks, show WCS is license-locked to Fernald waste, show Lynas Seadrift died on exactly this, and give you the intra-sector pair (own-your-sink MP vs no-sink processors) with the trigger dates to watch.' To EnergySolutions/Robuck: 'A paid advisory engagement to size and time the REE source-material-tails demand wave hitting Clive, and to position your license envelope to capture it before competitors realize they need you.' To OSC: 'An independent pre-loan constraint briefing so you do not finance a Dy/Tb plant that has no licensed place to put its tails for 30 years.'

FIRST MOVE (THIS WEEK)

This week: send a one-page dated memo to a named materials/critical-minerals PM at a resources-exposed fund, leading with the single sharpest line ('the gate is the tails sink, not separation: only EnergySolutions Clive and WCS Andrews can take it, WCS is locked to Fernald waste, and Lynas Seadrift already died on this'), and offer the full memo + 30-min call for a fixed fee. In parallel, email Ken Robuck at EnergySolutions with the demand-wave hook.

CHECKPOINTS

By end of 2026: watch whether WCS (Andrews) or EnergySolutions (Clive) files to amend its byproduct/source-material license to broaden acceptance beyond Fernald-type waste toward REE tails. A filing confirms the gate is binding and being chased. | Mid-2026: MP Materials' Dy/Tb circuit commissioning at Mountain Pass. If it runs on its own zero-discharge lined sink while competitors stall on residue permits, the own-your-sink thesis is validated. | Through 2026: any new OSC/DPA Title III REE-separation loan that does NOT name a licensed tails-disposal path is a sign the constraint is still unpriced and the intel is still sellable; a loan that DOES require it is a sign consensus is catching up (move faster or the edge closes).

DISQUALIFIER

Drop it if WCS or EnergySolutions broadens its license and visibly stands up ample REE-tails capacity (the sink stops being scarce), OR if the NRC walks back the 0.05% source-material trigger so HREE feeds escape 10 CFR 40 (the gate disappears), OR if a fund desk reads the memo and says 'already in our model' (consensus has caught up and the intel no longer commands a premium).

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The intel product is unsellable as written because its two load-bearing facts are wrong, and the wrongness is exactly what a fund desk checks first. (1) The plan's spine ("NRC/IAEA now treat any feed over 0.05% Th/U as source material, dragging every Western Dy/Tb plant into 10 CFR 40") misstates the rule. 0.05% is the decades-old statutory ORE definition, not a new 2025/2026 reclassification, and 10 CFR 40.13(c)(1)(vi) gives rare earths a specific exemption up to 0.25% Th/U for metals/compounds/products. There is no dated reclassification trigger; the "timestamped causal chain" artifact is fabricated. (2) Lynas Seadrift did not die on a missing 11e(2) radioactive-tails SINK. It stalled because Dow withdrew its TCEQ WASTEWATER discharge permit amendment in Feb 2025 after Diane Wilson's challenge (a liquid-effluent / Dow plastics-outfall fight), a different chokepoint than solid source-material tails disposal. The single sharpest line the plan leads with to a PM is built on two misattributions, and a materials/nuclear-licensing analyst refutes it in the first read; non-consensus here partly means wrong. Even setting the facts aside, the dollar still does not arrive: a resources PM at a multi-strat buys expert-network calls (GLG/Tegus, hundreds to low-thousands per call) and pays established shops on a relationship, not $15-40k one-off plus $5-15k/mo to a cold, unknown solo shop. EnergySolutions is an ECP-owned firm with its own commercial/BD team and will not pay a solo forecaster $10-25k/mo to size its own pipeline. OSC pays nothing (plan concedes this). And the only tradable expression, long MP vs short no-sink processors, is the already-crowded "MP is the only US game / MP owns its sink" narrative that sits in every MP bull deck, with no clean short leg (the Lynas catalyst already fired in Feb 2025 and is priced).

FIX OR KILL

Fix the facts first or the product is dead on arrival: drop the fake "new 0.05% reclassification" trigger, state the real rule (0.05% ore definition + the 0.25% rare-earth 40.13(c)(1)(vi) exemption, with the open question of where in the flowsheet the exemption stops applying and waste/incoming-ore streams get pulled into 10 CFR 40), and re-cast Lynas Seadrift correctly as a wastewater-discharge/permitting death, not a tails-sink death. The honest, still-interesting thesis becomes broader: residue + effluent + source-material permitting collectively gate Western HREE, and few investors model the full permitting stack. Then sell it the way this spend actually flows: not a $15-40k memo from an unknown shop, but a single paid expert-network-style call (route through Tegus/GLG/Dialectica or a one-page teaser to a named materials PM) at $1-3k to establish the relationship and test whether the corrected read is non-consensus. Best-fit buyer is a fund that TRADES the REE complex (a critical-minerals-focused L/S or the materials book at a multi-strat), not an operator. Drop the EnergySolutions and OSC retainer legs entirely as paying customers; they are at best free reputation contacts. If, after correction, the first paid call comes back "already in our model" (likely, given Lynas coverage and the MP-owns-its-sink narrative), the edge has closed and it should die.

> HARDENED ASK
>
> To one named materials/critical-minerals PM at a fund that actively trades MP/USAR/Lynas: a single paid 30-60 minute expert call (or a $1-3k scoped note) on the full Western-HREE permitting stack, factually corrected, leading with the honest framing: the binding gates are residue disposal AND wastewater/effluent permitting (this is what actually stalled Lynas Seadrift in Feb 2025), with the two private commercial source-material sinks and the 0.25% rare-earth exemption boundary as the under-modeled detail. Sell the relationship and the corrected map, not a $15-40k memo. Operators (EnergySolutions, MP, OSC) are reputation contacts, not paying customers.

REALISTIC TICKET

One-off paid call / scoped note in the $1-3k range to test the relationship, not $15-40k; recurring intel feed only if a fund explicitly asks to keep it, at expert-network rates, not the claimed $5-15k/mo. EnergySolutions/OSC retainer revenue: zero.

F2/NF3 electrolytic cell capacity is the no-buffer fab-fluorine chokepoint; rent migrates above fluorspar/HF to qualified near-fab cell-fleet owners (Linde, Kanto Denka) and the carbon-anode maker beneath them (Toyo Tanso) as Mitsui exits and fab fluorine intensity rises.

PASS

WHY CAPTURABLE

This is capturable now, by two distinct mechanisms, because the chokepoint is real, dated, and already half-priced in one liquid name. Elemental F2 cannot be shipped or stored, so generation must sit at the fab and cell-fleet capacity is a hard flow constraint, not a buyable inventory. The dated crack is confirmed: Mitsui Chemicals exits NF3 by March 2026, leaving Kanto Denka (TSE:4047) at ~90% of Japan, and the Aug 2025 Kanto Denka fire already showed the market reacts violently (4047 ran roughly 796 to 3,775 JPY over the trailing year). That gives us a clean, liquid instrument to take a position ourselves and a credible, urgent reason for fabs and gas majors to pay for a constraint-intel feed. Our data layer is blind to this midstream (only generic mining rows matched, no concept_flow, no traded prediction market), which is itself the edge: the constraint is unpriced in the places crowds look and only partly priced in one equity. The deepest inelastic input is not even NF3 itself but the isotropic carbon anode for the electrolysis cell, where Toyo Tanso is the named dominant maker. The risk is timing (the call resolves 2032) and that the equity move is partly spent, so the position must be structured around the next binding event, not the thesis in general.

TARGETS

Linde plc (Linde Electronics / GENERATION F on-site fluorine business) Head of Electronics / GENERATION F on-site generation business line (find the named VP, Linde Electronics)

Why them: Linde's GENERATION F is the named industry-standard on-site F2 cell fleet sitting inside fabs (SK Hynix among references). If cell-fleet capacity becomes the gating constraint on fab ramp, Linde owns the rent: it can price qualified near-fab F2 cells as scarce flow capacity rather than a commodity gas service. They win biggest if the thesis is right and need a dated read on where fab

VALUE MECHANISM

Two real dollar paths. Primary: a position we take ourselves - long the liquid equity that prices the chokepoint (Kanto Denka, TSE:4047) and/or the anode pick-and-shovel (Toyo Tanso, TSE:5310), sized to the next binding catalyst not the 2032 horizon, with a defined kill. Secondary, non-dilutive: a paid constraint-intel feed - a retainer from a commodities/chip-supply fund for the dated read and trade calendar, and a quarterly subscription from an exposed operator (Linde Electronics, Merck/Versum) for the cell-qualification bottleneck map.

WHO PAYS

A hard-asset/commodities or chip-supply-chain fund (retainer ~$10-50k for the dated read + trade calendar); separately Linde Electronics and Merck/Versum (quarterly cell-qualification bottleneck map ~$5-20k/quarter). If we self-express: long 4047 and/or 5310 sized to the next catalyst.

OUR ANGLE

A small independent forecasting shop is credible here precisely because the value is a sharp dependency-graph read, not headcount: we name the real layer (point-of-use F2 electrolysis cells), the real names (Linde GENERATION F, Kanto Denka, Toyo Tanso anodes, Merck/Versum DOE grant), and the dated crack (Mitsui exit, Kanto fire), and we prove discipline with a leak-free scored record and explicit kill-criteria. Funds and operators buy the one thing a big consultancy will not give them: a falsifiable, pre-consensus call on where the rent moves next, with the trade attached.

PROOF TO SHOW

The Vaticinus dependency-graph read that walks above fluorspar/HF to the un-stockpilable F2 cell fleet, and then below it to the Toyo Tanso carbon anode - the chokepoint-beneath-the-chokepoint no one is pricing -

fluorine intensity and NF3 merchant withdrawal force the next qualification bottleneck.

Reach: Cold email to Linde Electronics business development plus a warm route via the specialty-gas trade press (gasworld) where they already comment; lead with the Mitsui-exit + Kanto-fire dated read.

Kanto Denka Kogyo Co., Ltd. (TSE:4047) `IR / Corporate Planning (the equity is the target, not a sale to them)`

Why them: Post-Mitsui sole-major Japanese NF3 supplier, ~90% share, direct lines to TSMC/Samsung/Micron/Kioxia/Rapidus, guiding sales +45% and profit +51%. The stock is the cleanest liquid expression of the chokepoint and already ran ~4-5x on it. We do not sell to them; we take a position and track the next binding event (fire-recovery capacity, capacity-expansion timing, qualification of imports from Korea).

Reach: No outreach needed; this is an instrument. Monitor IR disclosures, METI import-substitution moves, and plant-status news for the next catalyst.

Merck KGaA (Versum Materials, Texas) `Head of Electronics Specialty Gases / the DOE on-site NF3 generation program lead`

Why them: Merck/Versum took a US DOE ~$45M grant for on-site NF3 generation and serves Intel/TSMC/Samsung in North America. They are the Western reshoring answer to the Japan single-point-of-failure and the most exposed buyer of a forward read on F2/NF3 cell-capacity qualification timing and where the next constraint binds (carbon anodes, cell fleet, HF feed).

Reach: Cold email to Versum/Merck Electronics commercial + the DOE grant program contact; angle: independent dated constraint map of the post-Mitsui fluorine midstream to de-risk their capacity build.

Toyo Tanso Co., Ltd. (TSE:5310) `Equity / supply-chain intel target (carbon anode maker beneath the cell)`

Why them: The isotropic carbon anode is the inelastic input inside every F2/NF3 electrolytic cell; Toyo Tanso (with 3M) holds the landmark electrode patents and is the named dominant supplier. If cell capacity is the constraint, anode supply is the constraint beneath the constraint. This is the dependency-graph read no one is pricing: a second liquid instrument and the pick-and-shovel of the whole thesis.

paired with our leak-free dated scored record and the fact this thesis is genuinely unpriced where crowds look (no traded prediction market, our own entity graph blind to the midstream) while only partly priced in one equity that already ran ~4-5x on the Aug 2025 Kanto fire. Lead with the dated pre-consensus framing: we flagged the cell fleet, not the gas, as the binding flow constraint.

INSTRUMENT

Clean instrument exists: long TSE:4047 (Kanto Denka Kogyo) as the direct chokepoint expression and long TSE:5310 (Toyo Tanso) as the carbon-anode pick-and-shovel beneath the cell. Direction long, sized to the next binding catalyst with a stop if Korean import substitution or restored capacity visibly relieves the constraint.

Reach: No outreach; second instrument plus a named edge to lead with when pitching funds. Verify current anode market share and capacity as a first move.

THE ASK

Two parallel asks. (1) To a hard-asset / commodities or chip-supply-chain fund: a paid constraint-intel engagement - we deliver a dated, leak-free read that the F2/NF3 electrolytic cell fleet (not fluorspar or HF) is the binding fab-fluorine constraint, with the next catalyst calendar (Mitsui exit completion March 2026, Kanto capacity/qualification timing, anode supply at Toyo Tanso) and the trade expression: long Kanto Denka 4047 and/or Toyo Tanso 5310 around the next binding event, with kill-criteria. (2) To Linde Electronics and Merck/Versum: a quarterly constraint-map subscription - where fab fluorine intensity (3D NAND / gate-all-around) and merchant NF3 withdrawal force the next cell-qualification bottleneck, scored and updated, so they site and qualify cell capacity ahead of the bind rather than after a fire.

FIRST MOVE (THIS WEEK)

This week, write a two-page dated constraint memo: the F2/NF3 cell fleet is the no-buffer fab-fluorine chokepoint, with the catalyst calendar (Mitsui exit March 2026, Kanto fire-recovery and capacity timing, Toyo Tanso anode supply) and the long-4047 / long-5310 expression plus kill-criteria. Send it as the opener to Linde Electronics business development and to the Merck/Versum + DOE on-site NF3 generation program, and pull current TSE:4047 and TSE:5310 quotes to set the entry. In parallel, find the named decision-maker: the Head of Linde Electronics on-site generation and the Versum/Merck electronics specialty-gas commercial lead.

CHECKPOINTS

By end of March 2026, confirm Mitsui Chemicals NF3 production has actually ceased and watch whether Kanto Denka (4047) and NF3 spot tighten on the lost cushion - if yes, the cell-capacity constraint is binding on schedule. | By Q3 2026, track whether Kanto Denka's fire-damaged Shibukawa capacity is fully restored and whether METI's Korea-import substitution closes the gap - restored slack would soften the near-term trade. | Within 6 months, see whether Linde Electronics or Merck/Versum publicly frame cell-fleet / on-site generation capacity (not HF) as the qualification bottleneck - a named operator saying it is early confirmation of the 2032 headline.

DISQUALIFIER

Drop the near-term trade and the urgency pitch if Korean NF3 imports plus restored Kanto capacity and the Merck/Versum DOE-funded on-site line visibly rebuild a comfortable supply cushion through 2027, so cell-fleet capacity stops being a no-buffer flow constraint and reverts to a priced commodity-gas service. At that point the constraint has been relieved and the rent does not migrate up to cell owners on the timeline that makes anyone pay now.

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The two paid-intel paths fail and the self-position is half-spent. (1) No fund retains an unknown solo shop at $10-50k for a single-name thesis memo: they already buy this exact supply-chain detail through expert networks (Tegus/AlphaSense ~$75-1,000/call, GLG) direct from ex-Kanto/Mitsui/Linde/Versum operators, sharper and more attributable than an outsider reading filings. No evidence of a budget line for single-thesis retainers to unbranded shops. (2) Linde and Merck/Versum ARE the primary source on cell-qualification timing (they own GENERATION F and the DOE NF3 line); selling them a map of their own bottleneck is selling water to the well, with no budget line and ~zero cold-email conversion. (3) The self-position, the only real dollar, is partly spent: 4047 is ~3,560 JPY, up ~300% on the year, with the only fresh sell-side mark at 1,514 JPY (Hold, under half current price), so the chokepoint is largely priced in the one liquid name. The named disqualifier (Korea imports + restored Kanto capacity + DOE-funded Versum line) is an active funded process, not a tail risk.

FIX OR KILL

Kill both intel-sale asks; they assume access and a buyer behavior that does not exist for an unknown shop. Reframe as a position, not a product. Cheapest real path: (a) verify the load-bearing edge first, Toyo Tanso (5310) isotropic-anode market share and capacity, because that is the only genuinely unpriced layer; (b) if the anode-beneath-the-cell read survives, take a small self-position in 5310 sized to the next catalyst, with 4047 only as a catalyst-driven trade given it has already run 4-5x; (c) the sole external move that fits an unknown shop is publishing the dated anode read as a free credibility artifact so a fund discovers it and books a call, never a cold-emailed retainer or operator subscription. Better-fit buyer if any: a small/micro-cap Japan-equity or hard-asset fund that TRADES 5310/4047, reached warm via the published read, not Linde/Merck who are operators and the source.

> HARDENED ASK
>
> No retainer, no operator subscription. First move: spend one day verifying Toyo Tanso (TSE:5310) isotropic-carbon-anode share and capacity for F2/NF3 cells, the one unpriced claim. If it holds, take a small self-position in 5310 (anode pick-and-shovel) sized to the next binding catalyst with a hard kill if Korean import substitution or restored Kanto capacity relieves the constraint; treat 4047 as a catalyst-only trade since it already ran ~300%. Publish the dated anode-beneath-the-cell memo as a free credibility artifact; if a Japan-equity or hard-asset fund engages off it, the paid step is a single paid pilot read (~$2-5k), not a quarterly contract.

REALISTIC TICKET

Self-position only: a small, catalyst-sized long in TSE:5310 (and opportunistically 4047 on a pullback to the next binding event), with defined kill. Any external dollar is a one-off paid pilot read in the low single-thousands at best, not a $10-50k retainer or $5-20k/quarter subscription. Treat external revenue here as ~$0 until a published artifact pulls inbound.

Crucible-grade quartz from Spruce Pine is a single-source, single-use consumable that rations both AI wafers and solar; the capturable edge is selling the binding-constraint read to funds trading HPQ-exposed equities and to the Western fabs/solar makers who carry the risk unhedged.

PASS

WHY CAPTURABLE

The physical chokepoint is real and tightly held: Sibelco (private, Belgian) and The Quartz Corp (50/50 JV of Euronext-listed Imerys and private Norsk Mineral) supply over 75% of the inner-layer ppb-pure quartz, the consumable is destroyed every ingot pull, and demand scales 1:1 with both AI wafers and solar. The 2022 15% cut into a 22% price spike and the 2024 Helene shutdown are dated proof the constraint binds and is not yet structurally hedged. It is not priced in any prediction market, and the two incumbents are largely private, so the capturable value is not a clean single equity. It is (1) a paid constraint-forecasting feed to commodity/critical-minerals and semis-supply-chain funds who already trade Imerys and the ASX silica juniors but lack a rigorous read on when substitution actually lands, and (2) a bespoke supply-risk advisory to the most-exposed Western buyers (fabs and crystalline-silicon solar) who have a single-source consumable on their critical path and no second source. We can express a direct position ourselves via long Imerys / long the credible ASX alt-source name as a satellite, but the realer, faster dollar is intel + advisory because the names are private or thinly covered.

TARGETS

Imerys (Euronext Paris: NK) Head of Investor Relations / Strategy lead for the High Purity Minerals (Performance Minerals) division

Why them: Imerys owns 50% of The Quartz Corp, one of two sources of crucible-grade quartz. The market values Imerys as a diversified industrial-minerals company and almost completely under-prices the strategic optionality of being a 50% owner of a global semiconductor/solar chokepoint. They are the only listed pure-ish lever on this thesis, and a dated, falsifiable read on demand binding through 2032 helps them tell the HPQ story to their own investors.

VALUE MECHANISM

Two real channels. Primary: a paid intel/advisory retainer. A quarterly constraint-forecast subscription to one or two critical-minerals/commodities funds (the report tells them direction on Imerys vs ASX alt-source), plus a bespoke supply-risk advisory engagement with one exposed Western fab/solar/wafer buyer (a fixed-fee note refreshed semi-annually). Secondary, expressed by us directly: a small satellite long position, long Imerys (NK) as the only listed incumbent lever and long the single ASX junior we judge closest to genuine ppb crucible grade as the asymmetric substitution call, sized as a position not a business.

WHO PAYS

A critical-minerals / commodities fund pays for the intel feed: roughly a 25k to 60k USD one-quarter trial scaling to a 100k to 250k USD annual retainer if a call lands ahead of consensus. A Western fab/solar/wafer buyer pays for the bespoke supply-risk advisory: roughly a 10k to 50k USD per engagement (matches the DRW-type bespoke constraint-forecasting ticket in our plan). If we instead take the position ourselves: long Imerys (Euronext: NK) plus a small long in one ASX silica junior (DRX or VRX), sized as a satellite, with the kill being a credible ppb crucible-grade qualification from China/Australia.

OUR ANGLE

We are a binding-constraint forecasting shop with a dated, leak-free scored record. This thesis is exactly our lane: rent accrues to the inelastic input (ppb inner-layer quartz), the consumable is destroyed each pull so demand is mechanical, and the question that actually determines every trade is when the constraint binds and whether substitution lands, which is a calibrated, falsifiable forecast, not a sell-side narrative. We already dated the 2022 price-spike mechanism and the Helene

Reach: Cold email to Imerys IR with a one-page dated constraint read; warm path via a sell-side metals/mining analyst who covers NK. They publish IR contacts. First move: find the named HPQ/Performance Minerals strategy owner on LinkedIn.

A critical-minerals / commodities hedge fund (e.g. a desk at a firm like Orion Resource Partners, or a thematic book at a multi-strat such as Millennium/Citadel covering materials) `PM or analyst on the critical-minerals / industrials-supply-chain book`

Why them: They want a non-consensus, dated, scored view on whether the Chinese (Qinling/Altay) and Australian (ASX silica juniors) substitution actually reaches ppb crucible grade and at what date. That single judgment determines the direction of the trade: long the incumbent scarcity (Imerys) vs long the alt-source juniors vs short crucible-dependent margins. This is exactly a binding-constraint, is-it-priced question, which is our edge.

Reach: Warm intro via the DRW-type quant/trading network already mapped in our people plan; cold-email angle: lead with the scored leak-free record and the dated 2022 price-spike call. First move: identify the specific materials/critical-minerals PM at one named fund.

A leading Western wafer/ingot or crystalline-silicon solar maker exposed to single-source crucibles (e.g. a US/EU CZ silicon or PV ingot operation such as a Hemlock/REC-type or a fab-feeding wafer supplier) `VP Supply Chain / Head of Strategic Procurement / Chief Risk Officer`

Why them: They have a single-source, single-use consumable on the critical path with no qualified second source and a live demonstration (Helene 2024) that one weather event halts them. They will pay for an independent, dated forecast of when the constraint binds, how far substitution is from crucible grade, and which alt-source to second-qualify now. This is buy-vs-stockpile-vs-qualify decision support, not a research curiosity.

Reach: Conference path (SEMICON West, SNEC PV); cold email to the named procurement VP with the Helene case as the hook. First move: pick one named Western CZ-silicon or wafer operator and find its procurement decision-maker.

Australian Silica Quartz Group / Diatreme Resources (ASX:DRX) / VRX Silica (ASX:VRX)

preview, and the call sits in no prediction market, which is the pre-consensus signature.

PROOF TO SHOW

The dependency-graph read that isolates the inner-layer ppb quartz as the single binding node (not polysilicon, not the crucible maker), our leak-free Brier-scored structural record (calibration moved 0.248 to 0.212 under the gate), and the dated pre-consensus mechanism: the 2022 Spruce Pine 15% output cut to 22% HPQ price spike and the 2024 Hurricane Helene shutdown, each logged before they were widely framed as a chip-supply risk.

INSTRUMENT

Long Imerys (Euronext Paris: NK) as the only listed incumbent lever on crucible-grade quartz, paired with a small long in the single most credible ppb-grade ASX silica junior (Diatreme ASX:DRX or VRX Silica ASX:VRX) as the asymmetric substitution call. Direction: long scarcity now, long the alt-source as the re-rate option; the pair self-hedges the substitution risk.

Managing Director / CEO and Head of Corporate Development

Why them: These are the listed substitution call. If the constraint binds and Spruce Pine cannot scale fast enough, an alt-source that can actually hit ppb crucible grade gets re-rated hard. They want a credible, dated demand-binding thesis they can hand to investors and to potential offtake partners (the fabs above). We can both advise them and take a position in the name we judge most credible.

Reach: Direct: ASX juniors are highly reachable, MDs answer email. First move: assess which of the three is closest to genuine ppb crucible grade (not just 4N flour), then approach that one.

THE ASK

To the funds: "We sell you a dated, scored constraint feed on crucible-grade quartz. Specifically, a quarterly binding-date forecast (P plus interval plus kill-criterion) on (a) whether HPQ supply rations wafer/solar output before 2032 and (b) when, if ever, Chinese/Australian substitution reaches ppb crucible grade. We will tell you which way that points Imerys vs the ASX juniors. Trial it for one quarter at a fixed fee; if our call moves before consensus, we move to a retainer." To the exposed buyer: "Pay us for an independent supply-risk note on your single-source crucible consumable: when it binds, how far each alt-source is from your spec, and a second-qualification shortlist, refreshed twice a year."

FIRST MOVE (THIS WEEK)

This week, write the one-page dated constraint note: the 2022 (15% cut to 22% spike) and 2024 (Helene shutdown) tells, a P-and-interval forecast that HPQ rations wafer/solar output before 2032 with an explicit kill-criterion (a verified ppb crucible-grade second source at scale), and the Imerys-vs-ASX-junior direction read. Send it to Imerys IR and to one named critical-minerals fund PM, leading with our leak-free scored record.

CHECKPOINTS

By Q3 2026, check whether Sibelco's doubled Spruce Pine capacity and The Quartz Corp output have actually cleared the post-Helene logistics constraint; sustained tight HPQ pricing keeps the thesis open. | By end 2026, track whether any Chinese (Qinling/Altay) or ASX junior source publishes verified ppb (5N-6N) crucible-grade results, not just 4N flour; a real qualification at scale starts to close the thesis. | By mid-2027, watch the 2026-27 fab ramps (TSMC AZ, Samsung TX, Intel Ohio, Rapidus) plus solar wafer additions against crucible-grade supply; demand outrunning the doubled capacity confirms binding.

DISQUALIFIER

A verified, at-scale ppb crucible-grade second source (China Qinling/Altay or an ASX junior) qualifying into real puller supply chains before the constraint binds, or HPQ prices going slack despite the fab/solar ramp, either means substitution beat the chokepoint and the scarcity trade is dead.

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The advisory/intel-sale dollar does not arrive. (1) The payers are not budgeted buyers for THIS from an unknown shop. A critical-minerals/commodities fund that trades Imerys already pays incumbent channels (CRU, Benchmark Mineral Intelligence, Wood Mackenzie, GLG/expert networks) and runs its own analysts; a solo shop selling a quarterly single-commodity "binding-date forecast" at 25-60k is a cold pitch into an occupied category, not a line item, especially on a 1.4B-total market. (2) The exposed Western wafer/solar buyer runs this in-house: their procurement and risk team lived the Helene shutdown and knows Sibelco/Quartz Corp better than we do; the valuable deliverable (an actually qualified second source) is a multi-year materials-engineering effort, not a 10-50k forecast note. (3) The edge has decayed: the Spruce Pine chokepoint is now mainstream retail coverage (Tom's Hardware, Wall Street Watchdogs "10T resource boom," multiple Substacks since Helene 2024), so "pre-consensus" is false in practice; and the plan's own disqualifier is already triggering, China classified HPQ as a new mineral species in April 2025 and domestic HPQ sand has been trialed in crucibles meeting service life. Selling an "is it priced / will substitution land" read while the kill-criterion is actively firing is a hard sell to any sharp PM. (4) Access is asserted, not held: the "DRW-type warm intro" and named-PM path are aspirational. The only money path that survives needs no buyer to say yes: a position we take ourselves.

FIX OR KILL

Drop the intel-subscription and bespoke-advisory channels as the primary path; they fail on budget, incumbency, in-house substitution, and a decaying edge. The clean, landable path is the position we express ourselves: a small satellite pair, long Imerys (Euronext: NK) as the only listed incumbent lever, paired with a small long in whichever ASX silica junior (DRX or VRX) is genuinely closest to verified ppb crucible grade, sized as a position not a business. The pair self-hedges substitution risk and requires no buyer, no access we lack, and no claim of pre-consensus. If any advisory dollar is pursued at all, the only honest version is a single cheap paid pilot to ONE ASX junior (not a fund, not a fab): a dated, scored demand-binding note they can hand to investors and offtake partners. ASX junior MDs answer email, and they have a real reason to pay for third-party validation of their own re-rate story. That is the cheapest test of whether anyone pays before investing in the feed. The fund-feed idea is effectively dead because of incumbent intel providers plus the firing disqualifier.

> HARDENED ASK
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> First and cheapest test: take the position ourselves. Open a small satellite long Imerys (NK) + small long in the one ASX silica junior we judge closest to verified ppb crucible grade, kill = a verified at-scale ppb crucible-grade second source qualifying into real puller supply chains. Separately, run exactly one low-cost paid pilot: offer one ASX junior MD (DRX or VRX) a fixed, small fee (about 5-10k) for a dated, scored, third-party demand-binding note on crucible-grade HPQ they can show investors and potential offtake partners. Do NOT pitch funds a quarterly subscription, the incumbents and the firing disqualifier kill it.

REALISTIC TICKET

No reliable advisory ticket. Honest revised position: a small satellite pair (long Imerys NK + small long one ASX silica junior), sized as a position not a business, low-single-digit-percent of a book. Optional one-off ASX-junior pilot at roughly 5-10k, treated as a single test of willingness-to-pay, not recurring revenue. The 100-250k retainer is a fantasy ticket here.

The binding constraint on Western rare-earth separation is roughly two dozen veteran SX/hydromet commissioning engineers, not capital or permits. The orgs that wrote the checks (DoD, DOE, public operators) are exposed to a delay they have not priced. Sell them the talent-risk read.

PASS

WHY CAPTURABLE

There is a reachable capture path because the thesis is concrete, datable, and now hits balance-sheet owners. The US government took a $400M equity stake in MP Materials and is funding Ucore, USA Rare Earth and others through DPA Title III and the new DOE Office of Critical Minerals, all targeting 2027-2033 ramps simultaneously. Every one of those programs is underwritten on a capital-and-permits model that ignores the human-capital chokepoint our needle isolates: experienced solvent-extraction commissioning engineers, two dozen in the West versus thousands in China, a cohort that takes 10-15 years to build so the 2030-2035 supply is already fixed. That makes delay non-diversifiable across the portfolio: if it binds, multiple taxpayer-backed plants miss spec at once. The buyer is whoever owns that aggregate risk (DoD program office, a fund holding the listed names) plus the individual operators who each privately know they cannot staff. Our edge is that we can name, plant by plant, who is most exposed and rank the risk, which neither the recruiters nor the banks are doing as a forecast. The honest constraint: there is no liquid instrument that isolates the talent factor cleanly, so the primary dollar is a paid intel/advisory feed and a scored risk-ranking, with a listed-equity position as a secondary expression.

TARGETS

U.S. Department of War / DoD Manufacturing Capability Expansion & Investment Prioritization (MCEIP / IBAS), DPA Title III critical-minerals desk

Director, Industrial Base Analysis & Sustainment, critical minerals portfolio

Why them: DoD put a $400M equity stake in MP Materials and DPA Title III money into Ucore (Louisiana) and others, all underwritten on capital-and-permits milestones. If the SX/hydromet talent cohort binds, several of its funded

VALUE MECHANISM

Paid intel/advisory: a 6-week paid diagnostic (the dated talent-risk ranking) converting to a quarterly retainer feeding the buyer's milestone, clawback, and retention decisions. The dollar arrives as a fixed-fee diagnostic then a recurring research retainer, billed to the program office or operator corp-dev budget. Secondary expression for our own book: a long-dated listed-equity pairs position (long the operator locking the cohort, underweight/short the most exposed) sized small, since no instrument isolates the talent factor cleanly.

WHO PAYS

Primary: a DoD program office (MCEIP/DPA Title III) or a public operator's corp-dev/strategy budget pays for the diagnostic and retainer. Diagnostic ticket roughly $25k-60k; standing retainer roughly $10k-30k/quarter, in line with bespoke constraint-forecasting for funds noted in our GTM. Secondary: a strategic-metals fund pays a similar research-note fee, or we take the position ourselves as a small long-dated MP/USAR-vs-peer pairs trade on a 2027-2030 horizon.

OUR ANGLE

We are the only party treating "can you staff and commission it" as a falsifiable, dated, scored forecast rather than a vague workforce worry. Recruiters (NES Fircroft, Davalyn, Bedford) sell placements; banks model capex and permits; nobody is ranking which plant slips because of the talent chokepoint and putting a Brier score on it. Our dependency-graph method plus leak-free record lets us name the exposed players and the inelastic input, and commit to a public, resolvable call, which is exactly the credibility a program office or fund needs to act.

PROOF TO SHOW

Our leak-free dated scored record plus the dependency-graph read that isolated this needle: show that we plants slip or run off-spec at the same time, an unhedged portfolio risk it is not currently scoring. A leak-free, plant-by-plant talent-risk ranking is decision-grade for milestone design and clawback terms.

Reach: DPA Title III posts public announcements and program contacts at businessdefense.gov/ibr/mceip; route a one-pager through the published MCEIP/DPAI contact and via the Office of Critical Minerals Strategy. Warm path: McAllister & Quinn / Brownstein critical-minerals advisors who already broker these programs.

MP Materials (NYSE: MP) `Chief Operating Officer / Head of Heavy Rare Earth Separation Commissioning - Michael Rosenthal (co-founder, COO)`

Why them: MP is commissioning heavy-REE separation at Mountain Pass imminently with a late-2026 full-separation target and a DoD/Apple-anchored balance sheet. Heavy separation (Dy/Tb) needs 1,000+ mixer-settler stages and the deepest veteran-operator expertise; a missed ramp is a public-market and national-security event. They privately know staffing is the risk; an outside, dated read on where the scarce talent is and who else is bidding for it is directly useful for retention and poaching defense.

Reach: Investor relations and corporate dev are reachable; better, a warm intro through a DoD/critical-minerals advisor or a mutual investor. Lead with the dependency-graph read, not a recruiting pitch.

USA Rare Earth, Inc. (NASDAQ: USAR) `Chief Operating Officer / VP Process Engineering (Stillwater OK + Colorado separation facility); commercial entry via Chief Commercial Officer - Chaitan Kansal (Chief Commercial Officer, appointed Apr 2026)`

Why them: USAR just commissioned a Colorado separation facility staffed by 28 engineers/scientists/technicians, with first-oxide targeted Q3 2026, and took DOE Critical Materials Innovation funding plus a Commerce equity stake. They are the textbook case of the needle: a built plant whose ramp now depends on a tiny pool of SX commissioning talent. A risk-ranking that shows where they sit versus MP and Ucore in the bidding war for the same two dozen people is buy-grade intel.

Reach: Public company with active IR (J.B. Lowe, Head of IR) and a new CCO; cold-email the CCO with a dated, specific exposure read; escalate to COO/VP Process Engineering once engaged.

named the human-capital cohort (two dozen Western SX/hydromet commissioning engineers) as the binding constraint over plant, ore, reagent and permit, with a 2030-2035 resolution date and kill-criteria, before the trade press converged on it. Pair it with the entity-graph output that surfaces the real exposed players (CMOC/State Grid on the China side; MP/USAR/Lynas/Ucore on the Western side) and one already-resolved Brier-scored call from the record to prove calibration.

INSTRUMENT

Long-dated listed-equity pairs: long the operator that demonstrably locks the scarce SX cohort, underweight or pair-short the most commissioning-exposed (MP Materials NYSE: MP and USA Rare Earth NASDAQ: USAR are the liquid handles), 2027-2030 horizon. Caveat: no instrument isolates the talent factor cleanly, so the primary value is the intel/advisory relationship; the trade is a secondary, small expression.

A critical-minerals-exposed fund or family office holding MP / USAR / Lynas (DRW-type quant or a thematic resources fund) `PM / analyst on the critical-minerals or strategic-metals book`

Why them: The talent-binding thesis is a dated, pre-consensus catalyst that the street is pricing as a capital-and-permits story. A fund can express our read directly: long the operator that locks the scarce cohort, underweight or pair-short the one most exposed to a commissioning miss, on a 2027-2030 window. They pay for a differentiated, falsifiable structural read with explicit kill-criteria.

Reach: Our existing DRW-type / resources-fund relationships and the fundraising network; pitch as a paid research note plus a scored call we will resolve publicly.

THE ASK

To the DoD MCEIP/DPA Title III critical-minerals desk: "We will deliver a dated, leak-free, plant-by-plant talent-risk ranking of every DoD- and DOE-funded Western separation project, scoring each on the one input you are not tracking: access to experienced SX/hydromet commissioning engineers. We will name who is most likely to slip or run off-spec by 2030 because they cannot staff, with explicit kill-criteria and a public Brier score at resolution. Start with a paid 6-week diagnostic for one program office; if it lands, a standing quarterly intel retainer feeds your milestone and clawback design." To the operators (MP, USAR): same ranking, reframed as competitive intelligence on the bidding war for the two dozen people.

FIRST MOVE (THIS WEEK)

This week: send a one-page dated exposure brief to USA Rare Earth's CCO Chaitan Kansal (public, newest, most reachable, textbook-exposed) and in parallel pull the named DPA Title III / MCEIP program contact from businessdefense.gov/ibr/mceip and email a two-paragraph version framing the cross-portfolio talent-binding risk. The brief leads with the named-entity read (MP, USAR, Lynas, Ucore) and one falsifiable, kill-criteria'd call.

CHECKPOINTS

By Q3 2026, watch whether USA Rare Earth hits its first-oxide target on schedule; a slip with any staffing/commissioning language in the 8-K or earnings call confirms the needle is already binding and sharpens the pitch. | By end-2026, watch MP Materials' heavy-REE separation commissioning updates; if full separation slips past late-2026 with capability/throughput caveats, the thesis is live and the operator is a hotter buyer. | Within 60 days, a reply (or hard no) from the DPA Title III/MCEIP contact or USAR's CCO tells us whether a program office/operator will pay for the talent-risk read; silence from both after two touches is an early close signal.

DISQUALIFIER

Drop this if a major automated or modular SX advance (continuous-counter-current or ML-tuned mixer-settler control) gets validated at commercial scale and removes the veteran-operator dependency, OR if it emerges that the funders are already buying this exact talent-risk ranking from an incumbent (a recruiter-advisory or a bank desk) such that our scored-forecast angle has no open lane.

CONFIDENCE

medium

ADVERSARIAL MONEY-PATH TEST - DOES NOT HOLD AS STATED

STRONGEST REFUTATION

The primary dollar (a paid DoD diagnostic-to-retainer) is the weakest link and it does not arrive as written. The DPA Title III program lives inside AFRL's Materials and Manufacturing Directorate at Wright-Patterson, which already staffs program managers, scientists, engineers, SMEs, AND incumbent support contractors plus its own contracting shop. A program office does not respond to a cold one-pager by cutting a $25k-60k purchase order to an unknown solo shop. Analytic support is bought via FAR contract vehicles, OTA consortia, or existing IDIQ support contractors (the McAllister & Quinn / Brownstein lobbyists in the plan broker grants TO recipients, they are not a buy-side procurement path for selling a research note). Title III dollars flow OUT as FOAs and cooperative agreements to plant builders, not to advisory vendors. So the named DoD buyer has no discretionary line item and no reachable mechanism for this purchase at this ticket from this seller. On the operator side (MP, USAR), the second problem: they already privately know staffing is the risk (the plan concedes this), so an outside ranking of who else is bidding for the same two dozen people is information they get for free from their own recruiters and their board, and a public Brier-scored call that names them as the plant most likely to slip is actively hostile to a public company managing its share price. They will not pay to be ranked as the loser, and they will not feed a stranger their retention strategy. Third, the edge is thinner than claimed: the talent shortage is already in the trade press (Reuters-tier coverage, CSIS, sector hiring trackers in 2025-26), so the pre-consensus window the proof rests on is largely closed. What is left, a falsifiable Brier-scored plant-by-plant ranking, is real but is a research-credibility artifact, not a thing either named buyer has budget to purchase.

FIX OR KILL

The intel-sale path to DoD and to operators is dead as a primary dollar (no budget line, no reachable vehicle, operators are adversarial to being publicly ranked). The salvageable expression is a different buyer and a different instrument. Better-fit buyer: a fund that TRADES the complex, not an operator that runs it. A strategic-metals or event-driven desk (the DRW-type relationship in the GTM, or a thematic resources fund holding MP/USAR/Lynas) is a real buyer of a differentiated, falsifiable, dated structural call with kill-criteria, because they can express it directly in a position and they do not care that the call is hostile to a name they can short. Cheaper pilot: a single paid research note ($5k-10k, one-off) to one such desk, resolved publicly, before pitching any retainer. Cleaner still given there is NO instrument that isolates the talent factor: take the position yourself, small, as the proof. A long-dated MP/USAR-vs-peer pairs expression sized as a tiny conviction position turns the forecast into a self-funded, public, Brier-and-P&L-scored track record, which is the actual asset, rather than chasing a government purchase order that the procurement structure does not allow. The honest move: kill the DoD/operator intel-sale framing, keep the call as a sold research note to a trading desk and/or a small self-held pairs position, and only then, if a desk pays twice, talk retainer.

> HARDENED ASK
>
> Do not pitch DoD or the operators. To one strategic-metals / event-driven PM (a DRW-type or thematic resources desk in the existing network): "Here is a dated, leak-free, plant-by-plant ranking of Western rare-

> earth separation projects scored on the one input the street prices as capital-and-permits, access to the ~two dozen veteran SX/hydromet commissioning engineers, naming who is most likely to miss spec or slip by 2030, with explicit kill-criteria and a public Brier score at resolution. One-off note for $5k-10k; if it resolves your way, a second note next quarter." In parallel, take the secondary expression ourselves: a small long-dated MP/USAR-vs-peer pairs position on a 2027-2030 window as self-funded, public proof.

REALISTIC TICKET

Primary intel-to-DoD/operator path: ~$0, it does not transact. Realistic alternative: a single $5k-10k one-off research note to a strategic-metals trading desk (not the $25k-60k diagnostic plus $10k-30k/qtr retainer claimed), plus a self-held pairs position sized small (low single-digit thousands of risk) as the real, controllable expression.